



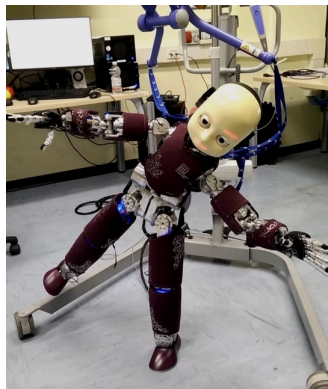
A Spectral-Grassman Wasserstein metric for operator representation of dynamical systems

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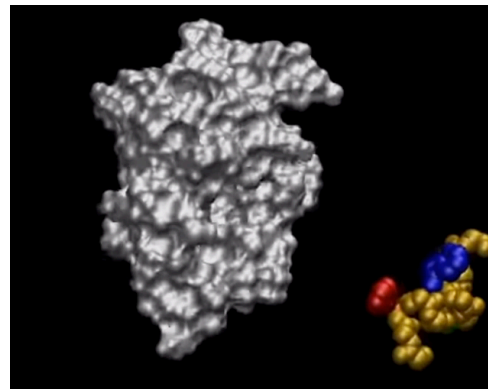
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1. Learning and interpreting dynamical systems
2. A Wasserstein metric on dynamical systems (SGOT)
3. Experiments

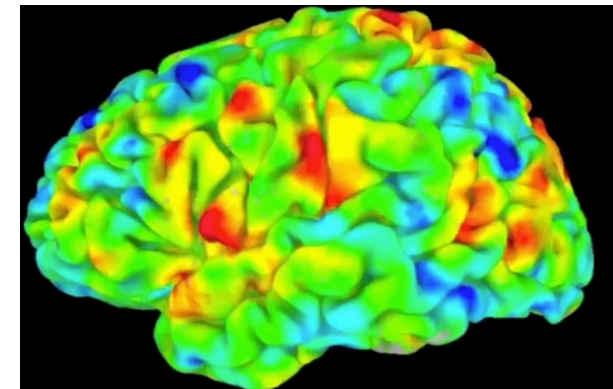
Dynamical systems are backbone models of temporally evolving phenomena:



Robotics



Molecular kinetic



Brain dynamic

Paradigm shift:

- Classical approach: ODE/PDE/SDE design + parameter fitting
- ML approach : Can we build dynamical models purely from observed data ?

➡Elegant formulation through transfer operator theory.

Let assume:

- A stochastic process: $(X_t)_{t \geq 0} \subseteq \mathcal{X}$
- A real-valued functional space: $\mathcal{F} \subset \mathbb{R}^{\mathcal{X}}$

Under some additional general assumptions, for any $t \geq 0$, there exists a linear operator $A_t : \mathcal{F} \rightarrow \mathcal{F}$, called *transfer operator*, such that:

$$[A_t(f)](x) \triangleq \mathbb{E}[f(X_t) | X_0 = x], \quad x \in \mathcal{X} \text{ and } f \in \mathcal{F}.$$

Remark: Even if the flow is dictated by a non-linear system, the transfert operator is linear.

Time-homogeneous systems: Since $A_{t+s} = A_t A_s$:

- (discrete): $A_t = A_1^t, \quad \forall t \in \mathbb{N}$
- (continuous): $A_t = \exp(tL), \quad \text{with } L = \lim_{t \rightarrow 0} (A_t - Id)/t \quad (\text{infinitesimal generator})$
 - ➔ It suffices to estimate $A = A_1$ or L .

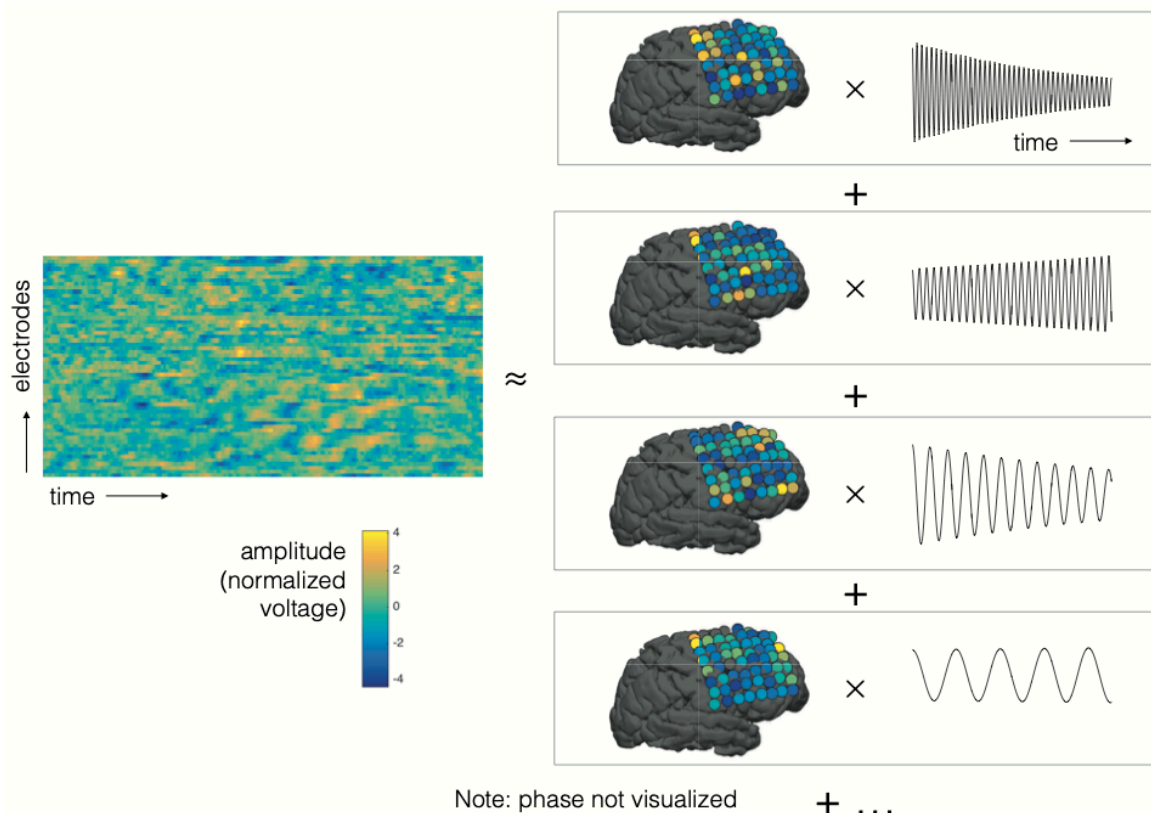
Assuming that $\mathcal{F}$ is a separable Hilbert space (typically $L^2_\pi(\mathcal{X})$ with π the system's invariant measure) and L is a non defective operator with purely discrete spectrum, then:

$$L = \sum_{j \in \mathbb{N}} \lambda_j g_j \otimes f_j, \quad \text{s.t.} \quad Lf_j = \lambda_j f_j, \quad L^*g_j = \bar{\lambda}_j g_j, \quad \langle f_i, g_j \rangle = \delta_{i,j}$$

It allows to decouple the evolution of any arbitrary observable $f \in \mathcal{F}$:

$$[A_t f](x) = \mathbb{E}[f(X_t) | X_0 = x] = \sum_{j \in \mathbb{N}} \boxed{e^{\lambda_j t}} \boxed{\langle g_j, f \rangle f_j(x)} = \sum_{j \in \mathbb{N}} \boxed{e^{\tau_j t + i2\pi\omega_j t}} \boxed{m_j^f(x)}$$

It disentangles operators into **temporal** and **static** components.



- $\tau_j = \mathcal{R}(\lambda_j)$ are dynamics' **timescales**.
- $\omega_j = \mathcal{I}(\lambda_j)/2\pi$ are **oscillating frequencies**.
- $m_j^f = \langle g_j, f \rangle f_j$ are Koopman modes of f .

Problem: Dynamical systems are only observed, and neither the operator A nor its domain $\mathcal{F}$ are known.

Popular approach [Bru22, Kos22]: Learning the action of $A : \mathcal{F} \rightarrow \mathcal{F}$ on a predefined RKHS $\mathcal{H} \subset \mathcal{F}$ with kernel $\kappa(x, y) \triangleq \langle \phi(x), \phi(y) \rangle$ via empirical risk minimization of the projected operator $P_{\mathcal{H}} A|_{\mathcal{H}}$, i.e.:

$$\operatorname{argmin}_{G \in HS(\mathcal{X})} \frac{1}{n} \sum_{i \in [n]} \|\phi(y_i) - G^* \phi(x_i)\|_{\mathcal{H}}^2$$

with the observation dataset $\{(y_i, x_i)\}_{i \in [n]}$ (typically a trajectory $y_i = x_{i+1}$).

➡ Under kernel universality assumption such methods come with spectral estimation guarantees [Kos23].

Alternative: Learning a representative finite dimensional RKHS $\mathcal{H}$ [Kos23].

While the estimation of individual dynamical systems has been thoroughly explored, theoretically grounded and computationally efficient methods for comparing different dynamical systems remain scarce.

Related work:

- Martin distance [Mar02] : pseudo-metric on ARMA models
- Binet-Cauchy kernel [Vid13]: Martin distance extension to LDS
- Topological conjugacy [Red24]: Optimal Transport on spectrum
- Hilbert-Schmidt and operator norms are too conservative

➡ Introduce Spectral-Grassmann Wasserstein Metric (SGOT)

Intuition:

Back to spectral decomposition:

$$L = \sum_{i \in \mathbb{N}} \sum_{j \in [n_i]} \lambda_i g_{i,j} \otimes f_{i,j} = \sum_{i \in \mathbb{N}} \lambda_i P_i, \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)}$$

➡ Spectral decomposition are unique up to a permutation (optimal transport) and a change of basis (Grassmann manifold) for spectral projectors (P_i).

Assumptions:

1. **(Time homogeneous dynamical systems):** Let consider $\{L_k\}_{k \in [N]}$.
2. **(Low rank operators):** For any $k \in [N]$ there exists $r_k \in \mathbb{N}$ such that the r_k eigenvalues the closest to the origin are separated from all the others.
3. **(A common functional space):** There exists an RKHS $\mathcal{H}$ such that for any operator, the estimation of its r_k -restriction, $T_k = \exp(L_k|_{r_k})$, is well defined.

Theorem: Let $\mathcal{H}$ be a separable $\mathbb{C}$ -Hilbert space and $\mathcal{S}_r(\mathcal{H})$ the set of non-defective operator on $\mathcal{H}$ with rank at most $r \in \mathbb{N}$. Let $(\mathcal{G}, d_{\mathcal{G}})$ the Grassmann manifold on the space Hilbert-Schmidt operators on $\mathcal{H}$. For any $\eta \in (0,1)$, let the embedding μ and the elementary metric d_{η} be defined as:

$$\mu(T) \triangleq \sum_{i \in \ell} \frac{m_i}{m_{tot}} \delta_{(\lambda_i, \mathcal{V}_i)} \quad \text{and} \quad d_{\eta}(T, T') \triangleq \eta |\lambda - \lambda'| + (1 - \eta) d_{\mathcal{G}}(\mathcal{V}, \mathcal{V}')$$

with $|\cdot|$ applied on polar coordinates of (λ, λ') , and $\mathcal{V}_i$ the m_i -dimensional subspace in $HS(\mathcal{H})$ spanned by the left/right eigenfunctions associated to the i st eigenvalue. Then $(\mathcal{S}_r(\mathcal{H}), d_{\mathcal{S}})$ is a metric space, where:

$$d_{\mathcal{S}}(T, T') \triangleq W_{d_{\eta}, p}(\mu(T), \mu(T'))$$

with $W_{d_{\eta}, p}$ the Wasserstein metric associated to d_{η} of order $p \in \mathbb{N}$.

Recall: Under kernel universality assumption such methods come with spectral estimation guarantees [Kos23], specifically for the Reduced Rank Regression (RRR) method [Kos22].

Theorem (simplified): Suppose two dynamical systems with low rank operator projection $T_1, T_2 \in \mathcal{S}_r(\mathcal{H})$ and $\hat{T}_1, \hat{T}_2$ their estimation from n samples with the RRR method [Kos22]. Suppose $\alpha \in (1, 2)$, and $\beta \in [0, 1]$ bounding the empirical covariance. In the i.i.d setting, for any $\delta \in (0, 1)$, with probability $1 - \delta$ it holds:

$$|d_{\mathcal{S}}(\hat{T}_1, \hat{T}_2) - d_{\mathcal{S}}(T_1, T_2)| \leq n^{-\frac{\alpha-1}{2(\alpha+\beta)}} \ln(2\delta^{-1})$$

Fréchet mean problem: Given $\{T_i\}_{i \in [N]}$ and $\gamma \in \mathbb{R}_+^N$ s.t. $\sum \gamma_i = 1$, we aim to solve:

$$\operatorname{argmin}_{T \in \mathcal{S}_r(\mathcal{H})} \sum_{i \in [N]} \gamma_i d_{\mathcal{S}}(T, T_i)^2$$

Parametric operator: Let consider:

$$T_{\theta} : h \in \mathcal{H} \mapsto \sum_{i \in [r]} \lambda_i \langle \kappa_x \alpha_i, h \rangle \kappa_x \beta_i \in \mathcal{H}$$

with $\theta = (\lambda, x, \alpha, \beta)$, $\lambda \in \mathbb{C}^r$, $x \in \mathcal{X}^n$ and $\alpha, \beta \in \mathbb{C}^{n \times r}$, $\kappa_x = \{\kappa(\cdot, x_j)\}_{j \in [n]}$ the set of representer functions such that $\kappa_x \alpha_i \triangleq \sum \kappa_{x_j} \alpha_{ji}$.

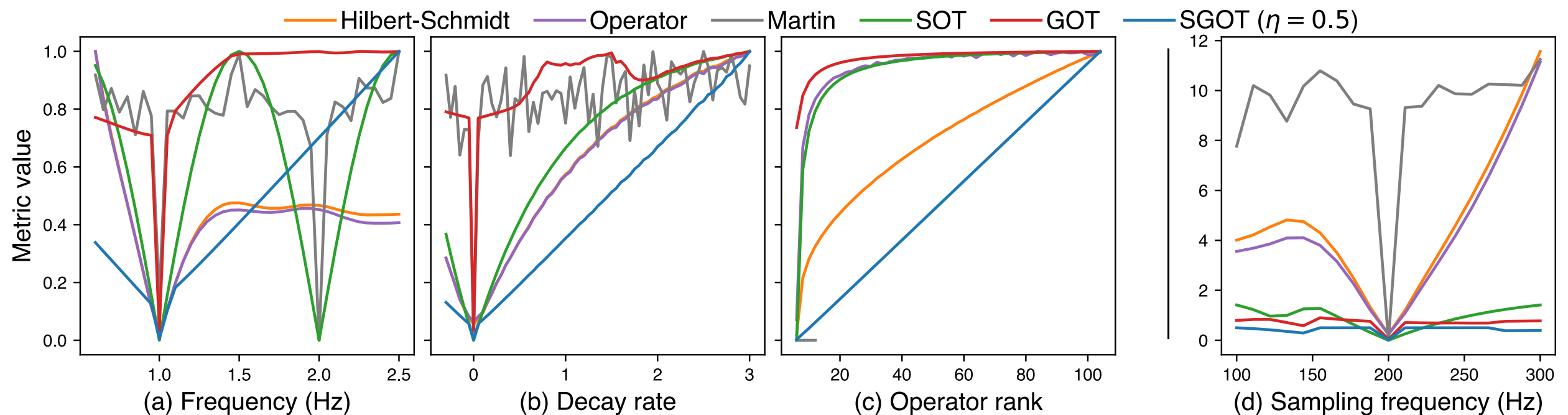
Constrained optimization problem:

$$\operatorname{argmin}_{P, \theta} \sum_{i \in [N]} \gamma_i \langle C_i(\theta), P_i \rangle \quad \text{s.t.} \quad \begin{cases} \alpha^* K \beta = I \\ \beta_j^* K \beta_j = 1, \forall j \in [N] \end{cases} \quad \begin{aligned} K &= \{\kappa(x_i, x_j)\}_{(i,j) \in [n]^2} \\ P_i &\in \Pi(\mu(T_{\theta}), \mu(T_i)), \forall i \in [N] \end{aligned}$$

Link to free-support barycenter, currently optimized with inexact block coordinates descend.

Metric comparisons between a baseline linear dynamical system and a shifted one under 4 scenarios:

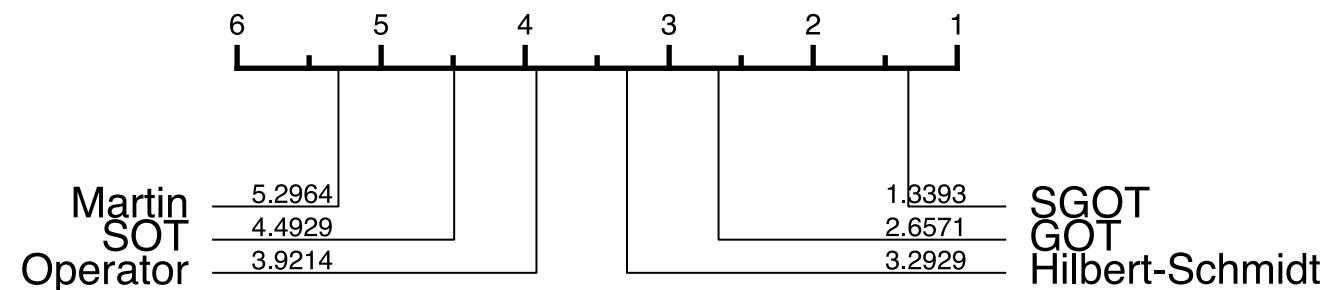
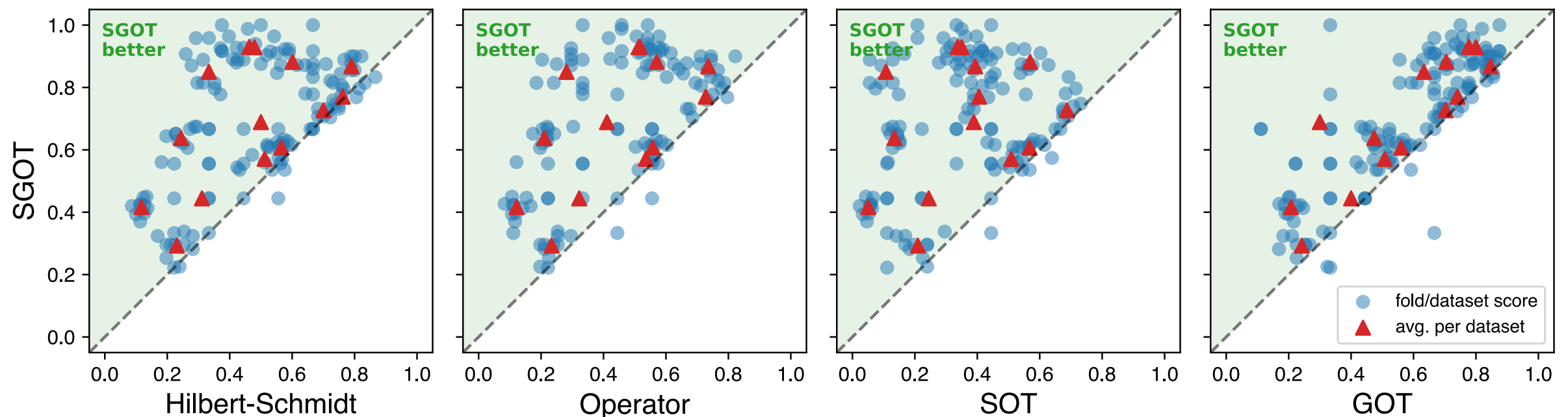
1. Frequency shift: left.
2. Decay shift: middle left.
3. Rank shift: middle right.
4. Sampling frequency shift: right.



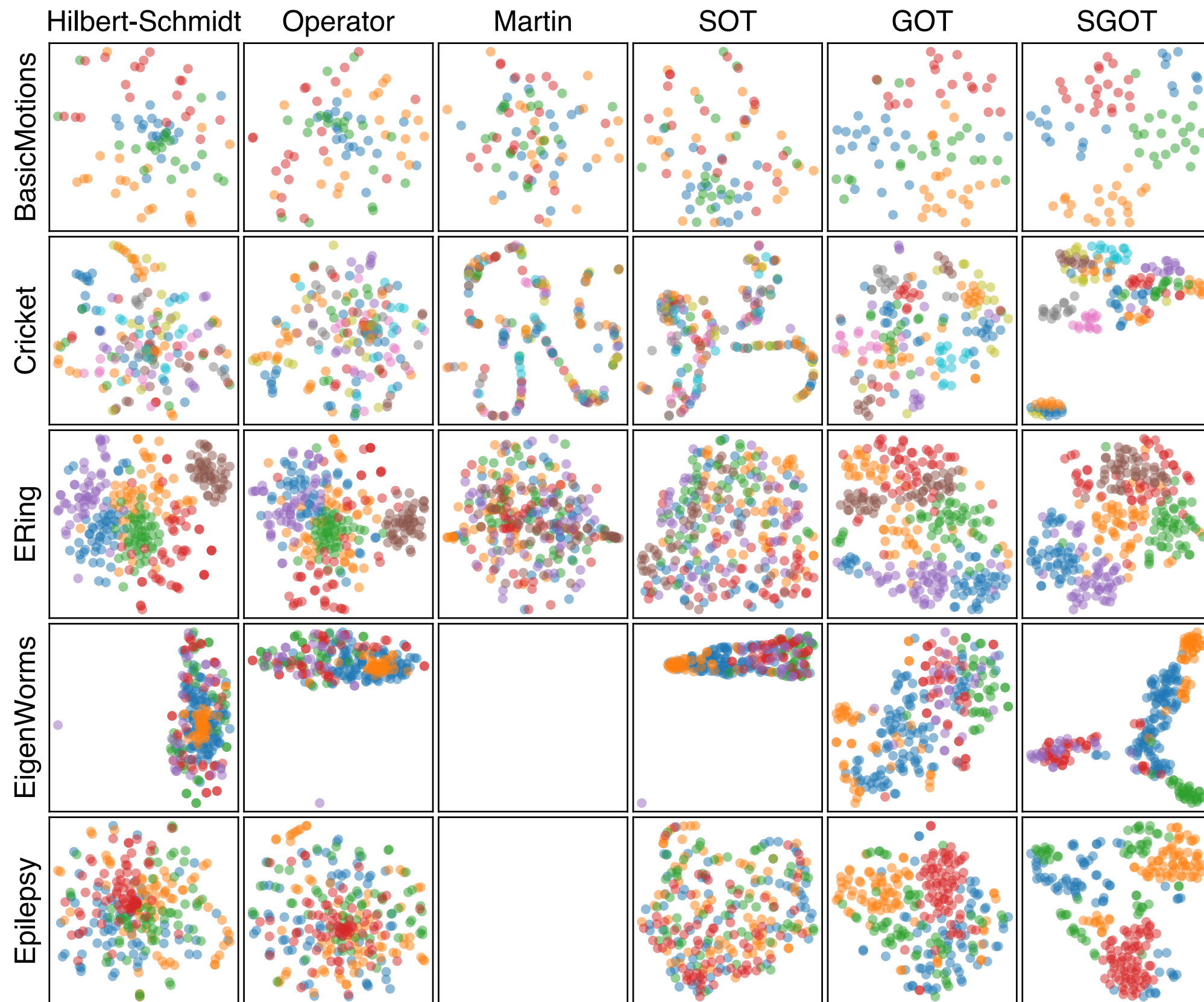
Dataset: 14 datasets from the UEA time series database.

Classifier: K-nearest-neighbors with 6 different metrics.

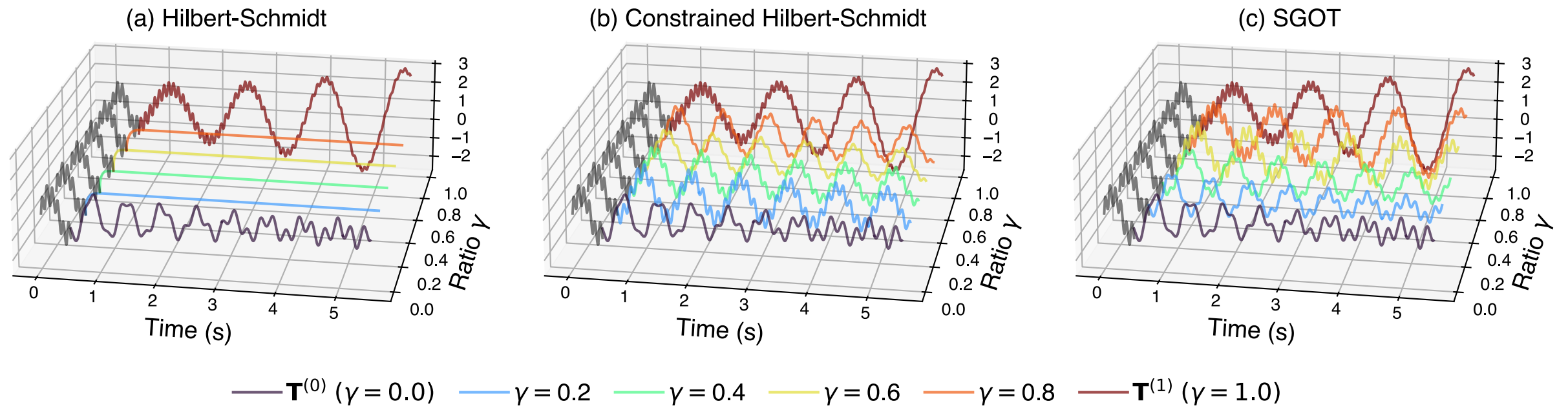
Score function: accuracy.



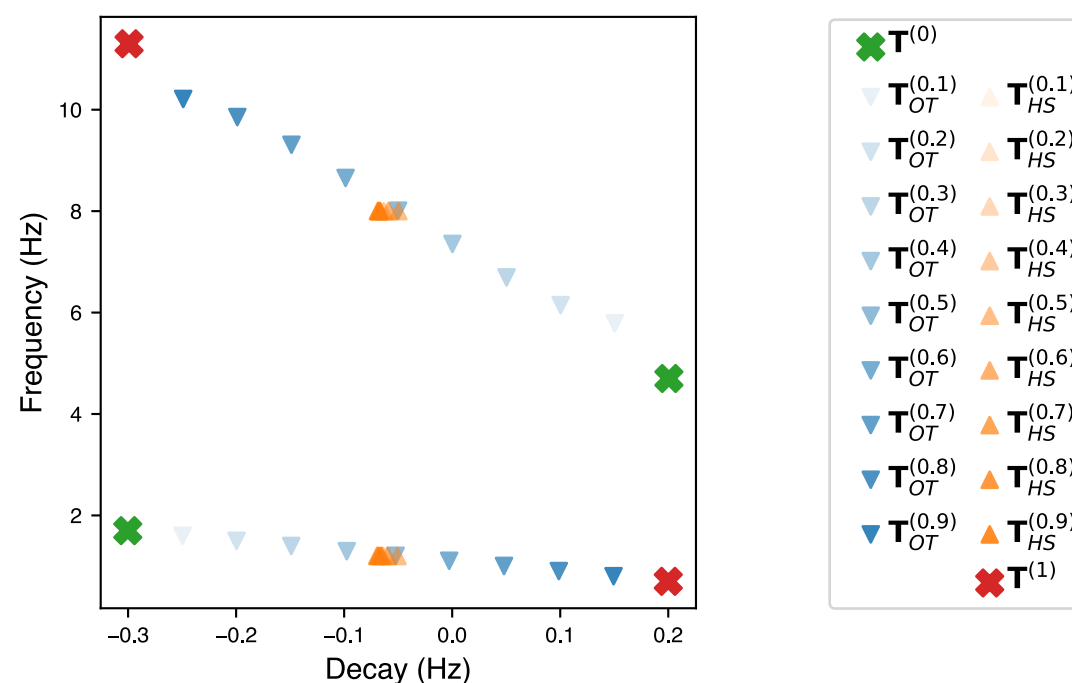
T-SNE embeddings



Barycentric interpolation between two linear dynamical systems.

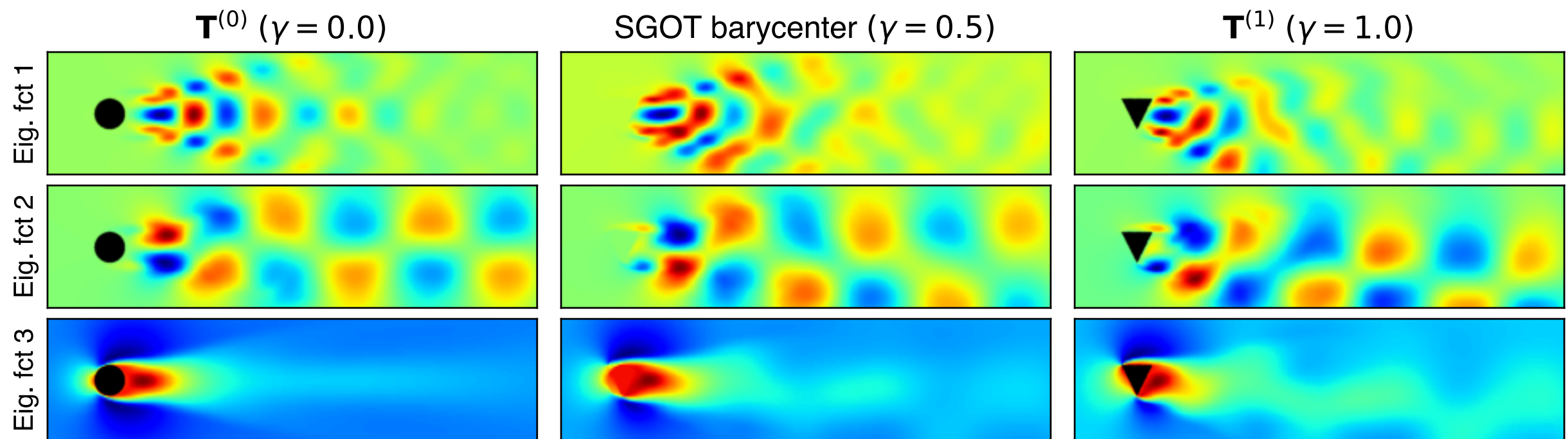


Barycenters' eigenvalue evolution:



Barycenter and interpolation

Barycentric interpolation between two fluid dynamics.



Thank you

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